Chapter 5 Supplement: Extended Screening & Mechanism Design

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Outline

1. Screening / Price Discrimination: Two-Type Case
2. Screening with Continuous Types
3. VCG Mechanism with Public Goods
4. Myerson's Optimal Mechanism
5. Wrap-Up and Exercises

Why Screening? A Mechanism Design Perspective

- **Screening** = a situation where the *uninformed* side (principal) offers a set of **contracts** to the *informed* side (agents).
- Agents self-select into the contract intended for their type *if* it satisfies IC, IR.
- In **price discrimination**, a monopolist doesn't know each consumer's valuation. The firm designs a menu (p, q) so each type **reveals** themselves by the choice of contract.

Relation to Mechanism Design

- Agents = consumers (types). - Mechanism = contract menu. - Must ensure no misreport (IC) and willingness to participate (IR).

Two-Type Model Setup

- Probability α a consumer is High type with valuation $\overline{\theta}$, and $(1-\alpha)$ Low type with $\underline{\theta} < \overline{\theta}$.
- A single product of "quality" (or quantity) q can be provided at cost $c(q)$ (often assume $c(q) = cq$ or something).
- The firm picks two contracts: (p_H, q_H) and (p_L, q_L).
- **Consumer Utility**: $U(\theta) = \theta\, q - p$ if they choose that contract, or 0 if they do not buy.

Goal

Maximize expected profit:

$$\max_{(p_H,q_H),(p_L,q_L)} \alpha\big[p_H - c(q_H)\big] + (1-\alpha)\big[p_L - c(q_L)\big],$$

subject to **IC** and **IR**.

Constraints: IR and IC

Individual Rationality:

$$\underbrace{\overline{\theta} q_H - p_H \geq 0}_{\text{IR-H}}, \quad \underbrace{\underline{\theta} q_L - p_L \geq 0}_{\text{IR-L}}.$$

If a type can get negative payoff from both contracts, they won't participate.

Incentive Compatibility:

$$\overline{\theta} q_H - p_H \;\geq\; \overline{\theta} q_L - p_L \quad \Longrightarrow \quad \text{(IC-H)},$$

$$\underline{\theta} q_L - p_L \;\geq\; \underline{\theta} q_H - p_H \quad \Longrightarrow \quad \text{(IC-L)}.$$

No Misreport

- IC-H: high type should prefer the "H" contract to "L." - IC-L: low type should prefer "L" contract to "H."

Typical Simplification of Constraints

A common result is:

- **IR-L** often binds: $\underline{\theta} q_L - p_L = 0$, so $p_L = \underline{\theta} q_L$.
- **IC-L** then becomes: $\underline{\theta} q_H - p_H \leq 0$. Often slack because of profit maximization.
- **IC-H** often binds.
- **IR-H** can be implied by IC-H plus IR-L, or might be slack if we can already ensure H gets positive surplus.

Key Trick

Often you find $q_L < q_H$ in optimum, $p_L < p_H$. Low type is served with lower quality to reduce the high type's incentive to "downgrade."

Deriving IR-H from IR-L + IC-H

IC-H:

$$\overline{\theta} q_H - p_H \geq \overline{\theta} q_L - p_L.$$

IR-L:

$$\underline{\theta} q_L - p_L \geq 0.$$

From IR-L, we get $p_L \leq \underline{\theta} q_L$. Substitute in IC-H:

$$\overline{\theta} q_H - p_H \ \geq \ \overline{\theta} q_L - \underline{\theta} q_L = (\overline{\theta} - \underline{\theta}) q_L \geq 0.$$

Solving the 2-Type Problem (Typical Outcome)

One typically finds:

- $q_L < q_H$,
- $p_L = \underline{\theta} q_L$ (L's IR binds),
- $p_H = \overline{\theta} q_H - (\overline{\theta} - \underline{\theta}) q_L$ (IC-H binding).
- So H obtains some surplus $(\overline{\theta} - \underline{\theta})(q_H - q_L)$ as "information rent," to keep them from choosing the L contract.

Result

Distortion: q_L might be depressed below the efficient level for the Low type, purely to discourage H from faking L. This is the hallmark of screening.

Interpretation for Price Discrimination

- q_H, p_H is the "premium version," q_L, p_L is the "basic version."
- The **low** type basically is served at a lesser quality to keep the menu incentive-compatible.
- The **high** type might pay more, but still enjoys some positive surplus if we can't fully extract it without losing them to the "low" contract.

Hence

We see how from a mechanism design viewpoint, the solution arises from IR-L, IC-H binding, ensuring no cross-type misreport.

Continuous Type Setup

Now let θ be drawn from a continuous distribution F on $[\theta_{\text{min}}, \theta_{\text{max}}]$. The firm does not observe θ but offers a continuum of contracts $(q(\hat{\theta}), p(\hat{\theta}))$. Consumer with true type θ picks $\hat{\theta}$ that maximizes

$$U(\theta, \hat{\theta}) = \theta \cdot q(\hat{\theta}) - p(\hat{\theta}).$$

IC Condition:

$$U(\theta, \theta) \geq U(\theta, \hat{\theta}), \quad \forall \hat{\theta}, \theta.$$

IR Condition:

$$U(\theta, \theta) \geq 0, \quad \forall \theta.$$

Envelope Result

- $$U(\theta, \hat{\theta}) = \theta \cdot q(\hat{\theta}) - p(\hat{\theta}).$$
- We get a classic envelope result:

$$\frac{d}{d\theta} U(\theta) = \frac{\partial}{\partial \hat{\theta}} [\theta q(\hat{\theta}) - p(\hat{\theta})] \Big|_{\hat{\theta}=\theta} = q(\theta).$$

Hence Envelope Theorem

$U(\theta) = U(\theta_{\min}) + \int_{\theta_{\min}}^{\theta} q(t)\, dt$. If $U(\theta_{\min}) = 0$ for IR at the bottom, that pins down $U(\theta)$ fully.

Monopolist's Profit Maximization

Firm's objective:

$$\int_{\theta_{\min}}^{\theta_{\max}} \Big[p(\theta) - c\big(q(\theta)\big)\Big] dF(\theta).$$

- Under IC, we know $p(\theta) = \theta \cdot q(\theta) - U(\theta)$.
- So profit $= \int \Big[\theta q(\theta) - U(\theta) - c(q(\theta))\Big] dF(\theta)$.
- Then we express $U(\theta)$ via the envelope approach: $U(\theta) = \int_{\theta_{\min}}^{\theta} q(t)dt$ (assuming $U(\theta_{\min}) = 0$).
- By Fubini's Theorem, we get that profit as

$$\int \Big\{[\theta q(\theta) - c(q(\theta))]f(\theta) - q(\theta)[1 - F(\theta)]\Big\} d\theta$$

We then optimize over $q(\theta)$

Potentially get $q(\theta) <$ the "efficient" quantity for lower θ. This is the continuous analog of "quality distortion for low type."

Conclusion on Continuous Screening

- We see the same phenomenon: **low types** might be served less than the efficient level, so that **high types** remain content with their contract and do not "pretend to be low."
- The final allocation is typically a downward distortion for all but the highest types, to extract surplus from top.

This Summarizes Mechanism Design Approach to Price Discrimination

Comparing to simpler "2-type" analysis, but extended to continuum of consumer valuations.

VCG Mechanism (Recall Brief)

Definition (repeat):

- n agents each with $v_i(o)$ over outcomes $o \in O$.
- Mechanism picks $o^* = \arg\max_o \sum_i \hat{v}_i(o)$.
- Payment $p_i = \max_o \sum_{j \neq i} \hat{v}_j(o) - \sum_{j \neq i} \hat{v}_j(o^*)$.

Key property: *Truth-telling* is a dominant strategy, ensures **efficient outcome**.

Now let's do a **public good** scenario to see the budget-balance problem.

Public Good Example

- n agents, each has private value v_i for a project costing C.
- If $\sum_i v_i \geq C$, the project is efficient to provide; otherwise, do not provide.
- In **VCG**:
  - Provide if $\sum_i \hat{v}_i \geq C$; otherwise not.
  - Payment for agent i: $p_i =$ "others' best alternative sum" $-$ "others' actual sum."

Detailed Payment Computation (Public Good)

If the project is **not** provided ($\sum \hat{v}_i < C$), then $p_i = 0$ for each i.

If **provided**, i.e. $\sum_i \hat{v}_i \geq C$:

$$p_i = \max_o \sum_{j \neq i} \hat{v}_j(o) \; - \; \sum_{j \neq i} \hat{v}_j(o^*).$$

But for "others alone," if agent i is absent, they might decide *not* to provide if $\sum_{j \neq i} \hat{v}_j < C$. So that sum is 0. $\sum_{j \neq i} \hat{v}_j(o^*)$ is the others' sum of valuations minus the cost of the project.

Budget Imbalance in VCG for Public Goods

- Summing all p_i can be lower than the cost of the project: the mechanism may have to *subsidize* the project from outside.
- This demonstrates that **VCG** ensures truthfulness + efficiency but not budget balance in many public good contexts.

Trade-off

No mechanism can achieve all of: (1) Efficiency, (2) IC/truth-telling, (3) Budget balance, (4) IR. The classic "Green–Laffont Impossibility." So we pick which conditions to relax.

Myerson's Optimal Mechanism: A Glimpse

- In **single-item** auctions with risk-neutral, i.i.d. valuations, we can go beyond "highest v wins" to *maximize revenue*.
- **Key Trick**: define $\phi(\theta) = \theta - \frac{1-F(\theta)}{f(\theta)} =$ "virtual valuation."
- **Allocate** to the bidder with highest $\phi(\theta)$ if it's positive, else don't sell. Payment is computed to keep IC.
- This mechanism may skip selling to types below some threshold. E.g. if F is uniform(0,1), $\phi(\theta) = 2\theta - 1$, so we need $\theta > 0.5$ to have a positive virtual value $\geq$ a "reserve price" of 0.5.

Connection to Screening

In screening, we *exclude* some low types if that boosts profit. Myerson's auction is the multi-bidder analog—only sell if $\phi(\theta)$ is positive.

Myerson's Optimal Mechanism: A Glimpse

- If multiple exceed r, choose the highest, paying some mechanism-based price.
- We can refine who gets the good based on distribution F to tweak expected revenue.
- That is the **optimal** approach under standard assumptions, overshadowing second-price if second-price doesn't pick a strong enough reserve or weighting.

Remark

A rigorous derivation uses the envelope theorem + IC in multi-bidder environment. The result is a "virtual value rule."

Wrap-Up So Far

- **Screening** (price discrimination) with 2-type or continuum: A mechanism design approach ensures each type chooses "their contract," giving us IR, IC constraints.
- **VCG revisited**: It's a general approach for efficiency, but *not* necessarily budget balanced, especially in public goods.
- **Myerson's Auction**: extends the logic of "exclusion of low types" to multi-bidder auctions for revenue maximization, using *virtual valuations*.

Common Thread

IC + IR appear in **all** these contexts, whether we talk about monopoly pricing, public goods, or auctions. Mechanism design unifies them.

Exercises (1): Screening, 2 Types

1. **Write the constraints**:
   - IR-L: $\underline{\theta} q_L - p_L \geq 0$, IR-H: $\overline{\theta} q_H - p_H \geq 0$,
   - IC-L: $\underline{\theta} q_L - p_L \geq \underline{\theta} q_H - p_H$,
   - IC-H: $\overline{\theta} q_H - p_H \geq \overline{\theta} q_L - p_L$.
2. **Max Profit**:

$$\max_{\{p_L, q_L, p_H, q_H\}} \alpha[p_H - c(q_H)] + (1-\alpha)[p_L - c(q_L)],$$

subject to those constraints. Show how IR-L and IC-H typically bind, letting you solve for p_H, p_L, q_H, q_L.

Exercises (2): Screening, Continuous

1. **Single-Crossing Condition**:
   - For θ in $[\theta_{\min}, \theta_{\max}]$, define $U(\theta) = \theta q(\theta) - p(\theta)$.
   - Show that $U'(\theta) = q(\theta)$ under IC.
2. **Monopolist's Problem**:
   - $\max \int [p(\theta) - c(q(\theta))] dF(\theta)$, with $p(\theta) = \theta q(\theta) - U(\theta)$.
   - Argue how $q(\theta)$ might be distorted downward for lower θ to reduce info-rent for higher θ.

Exercises (3): VCG Public Good

1. **Simple 3-agent case**:
   - If $\sum_i \hat{v}_i \geq C$, provide the public good, else not.
   - Write p_i for each agent.
   - Show how total $\sum_i p_i - C$ might be negative if an agent is pivotal.
2. **Discuss**:
   - Why is it IC & efficient, but not budget-balanced?

Exercises (4): Myerson's Mechanism

1. **Definition of Virtual Valuation**:

$$\phi(\theta) = \theta - \frac{1 - F(\theta)}{f(\theta)}.$$

   - Show how a uniform(0,1) distribution yields $\phi(\theta) = 2\theta - 1$.
   - That implies a "reserve" at $\theta = 0.5$ if $\phi(\theta)$ must be > 0 to sell.

2. **Connection to 2-type screening?**
   - If you had n bidders each with a 2-type distribution, how does Myerson's auction handle them?
   - Argue it's akin to "excluding" low types if that yields higher expected revenue.

References (1/2)

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References (2/2)

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- Clarke, E. (1971). *Multipart pricing of public goods*.
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- Myerson, R. (1981). *Optimal Auction Design*.

Concluding Remarks (1/2)

- **Screening** = principal's approach to handle unknown types (high vs. low) via a menu of contracts. Key constraints: IC, IR.
- **Two-type** model reveals how $q_L < q_H$ and certain constraints bind.
- **Continuous-type** model generalizes with single-crossing, envelope theorem for $u(\theta)$, typically leading to downward distortion of $q(\theta)$ except at top types.

Concluding Remarks (2/2)

- **VCG**: Another universal design for *efficiency*, but e.g. public goods often face budget imbalance.
- **Myerson's Auction**: revenue-centric design, skipping some lower types. This parallels screening logic: "exclude low θ if it doesn't pay off to serve them."
- Mechanism design connects **auctions, screening, public goods** under a single framework.

Thank you! Q&A?